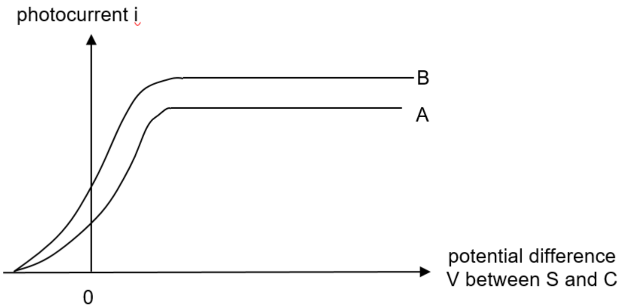
5	**(a)**		A photon is a quantum of electromagnetic radiation and the particulate representation of electromagnetic radiation that carries with it an amount of energy E = hf where h is the Planck constandt and f the frequency of the radiation.	B1 B1
	(b)		$e = \frac{hc}{\lambda} = \frac{(6.63\times10^{-34})(3\times10^{8})}{265\times10^{-9}}$ $= 7.51\times10^{-19}\ J$ = 4.69 eV	M1 A1
	(c)	**(i)**	The work function of a surface is the minimum energy of the photons in the incident radiation on the surface required to cause photoemission of electrons.	B1
		(ii)	The minimum potential difference V_s will become smaller. Light of higher wavelength has lower energy. Based on conservation of energy, when photons of lower energy are absorbed by electrons in the rhodium metal, some energy is used to release the electrons from the metal surface and the rest of the energy will determine the maximum KE of emitted photoelectrons. Since the stopping potential depends on the maximum KE of photoelectrons emitted which will be converted to electric potential energy as the photoelectrons travel towards the collector, the stopping potential needed is lower. OR Use of E = hf_o + eVs, higher wavelength has lower energy minimum potential difference V_s will become smaller	B1 B1 B1 (B1) (B1) (B1)
	(d)		If collector cup C is higher potential than the rhodium surface S, increasing V will only increase the kinetic energy of each photoelectrons reaching the collector cup C , and does not change the maximum rate of arrival of photoelectrons at C. The maximum rate of photoelectrons (emitted from S and so) reaching C is dependent on the intensity of the ultraviolet radiation directed at the rhodium surface. Hence, photoelectric current does not increase when V is made larger. Mentioning that it is the saturation current = only 1 mark	B1 B1

	(e)	 Graph is labelled as B Same V_s shown Higher i shown	 B1 B1